

EXPERIENCE

SDE Intern May — July 2025
Algoleap Technologies Pvt. Ltd. | [Certificate](#) Hyderabad

- Cut multimodal AI integration and rate-limit errors by 30% by architecting a unified GenAI API design spanning text, image, audio, and video generation.
- Eliminated the need for 4 separate API keys by designing a single consistent GenAI interface.

Intern Dec 2024 — Jan 2025
Indian Association for the Cultivation of Science | [Certificate](#) | [Website](#) Kolkata

- Boosted departmental operational efficiency by 40% by architecting and shipping a login-secured Store Inventory Job Card Portal for end-to-end job card tracking.
- Processed 8,000+ applications with zero downtime by designing a scalable platform architecture.

SKILLS, TECHNOLOGIES AND CERTIFICATIONS

Languages	HTML, CSS, JavaScript, C, C++, Java, Python, SQL
Frameworks & Architecture	Node.js, Express.js, REST API Design
Cloud and DevOps	AWS, Docker, CI/CD
Databases	MySQL, PostgreSQL, Redis, Neo4j, MongoDB
Certifications	AWS Certified Developer - Associate , AWS Certified Solutions Architect - Associate , AWS Certified Cloud Practitioner

PROJECTS

LiveLingo | Node.js, Express.js, AWS Interactive Video Service, Gemini Flash 3.1 Live Github

- Built a real-time multilingual speech pipeline translating English into Indic languages, engineering an end-to-end Node.js/FFmpeg middleware with WebSockets and the Gemini Flash 3.1 Live API to stream translated audio via AWS IVS.
- Achieved sub-400ms audio latency by streaming continuous 20ms PCM chunks, while evaluating output quality and stability under constrained network conditions (350 Kbps), quantifying TCP Head-of-Line blocking impacts (17.3s spike) and semantic drift across languages.

Validex | Node.js, Express.js, Azure Maps, Azure OCR, Twilio, Langgraph Github | Website | Demo Video

- Designed an agentic AI copilot for healthcare provider data validation, automating directory updates and reducing manual verification workloads by 80-90% — processing 100 provider records in under 5 minutes with 85%+ extraction accuracy from unstructured PDFs.
- Selected for EY Techathon 6.0 Semi Finale 2026 ([Certificate](#))

Memory Lens | Flutter, Dart, Node.js, Express.js, AWS Rekognition, Qdrant DB, OpenAI Embedding Github | Demo Video

- Built a voice-driven semantic image search system using NLP-based embeddings and vector search, achieving sub-2-second search latency across 1,000+ images with 95%+ voice recognition accuracy and 85%+ semantic relevance.
- Selected for Samsung PRISM GenAI Hackathon Semi Finale 2025 ([Certificate](#))

Conference Management Toolkit | Node.js, Express.js, Bootstrap, AWS EC2, RDS, Redis, Nodemailer Github | Website

- Engineered an end-to-end conference management toolkit (clone of [Microsoft CMT](#)) to automate paper and poster submissions, streamlining the entire academic review lifecycle for 100+ concurrent users.
- Implemented a robust server-side architecture featuring automated email notifications and dynamic request routing, reducing manual administrative processing time by 40% and ensuring 100% reliability in submission tracking.

PUBLICATIONS

A System and Method for Voice-Driven Semantic Image Search Leveraging AWS and OpenAI, V. Khurana, G. Satsangi, A. Sharma, Dr. Umamaheswari M - Accepted for presentation and publication, IEEE INCECT 2026 (Aug 21–22, 2026)

ACHIEVEMENTS

Honorable Mention Award , Bangkok University x Berkley SkyDeck Fund Hackathon 2026	Certificate
Best AI Model , Yantra Central Hackathon 2025, VIT Vellore	Certificate
Winner , Aquathon 2024 (Technical Hackathon on World Water Day), VIT Vellore	Certificate

EDUCATION

Bachelor of Technology in Computer Science Engineering, Vellore Institute of Technology, Vellore 2023 - 2027
CGPA: 8.43 / 10.00